

NUMERICAL INTEGRATION

- **Trapezoidal Rule:** $y = f(x)$ be a continuous function defined on $[a, b]$. Divide the interval on x-axis into n equal parts of length $h = \frac{b-a}{n}$. Let $a = x_0, x_1, x_2, \dots, x_n = b$ be the abscissa of successive points of division and $y_r = f(x_r)$ for $r = 0, 1, 2, \dots, n$ then:

$$\int_a^b f(x)dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n]$$

- **Simson's 1/3rd Rule:** $y = f(x)$ be a continuous function defined on $[a, b]$. Divide the interval on x-axis into $n = 2m(\text{even})$ equal parts of length $h = \frac{b-a}{n} = \frac{b-a}{2m}$. Let $a = x_0, x_1, x_2, \dots, x_n = b$ be the abscissa of successive points of division and $y_r = f(x_r)$ for $r = 0, 1, 2, \dots, n$ then:

$$\int_a^b f(x)dx = \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \dots) + 2(y_2 + y_4 + y_6 + \dots) + y_n]$$

Simpon's Rule is more accurate than the Trapezoidal Rule.